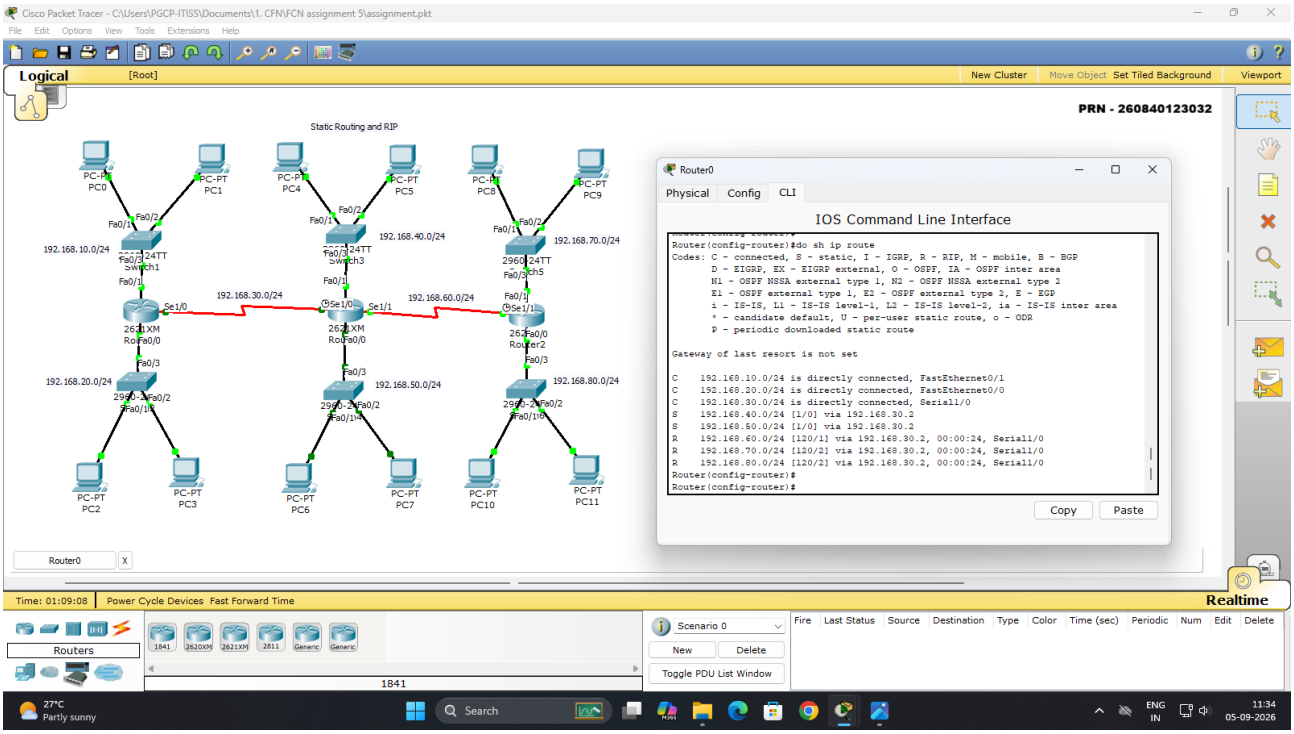
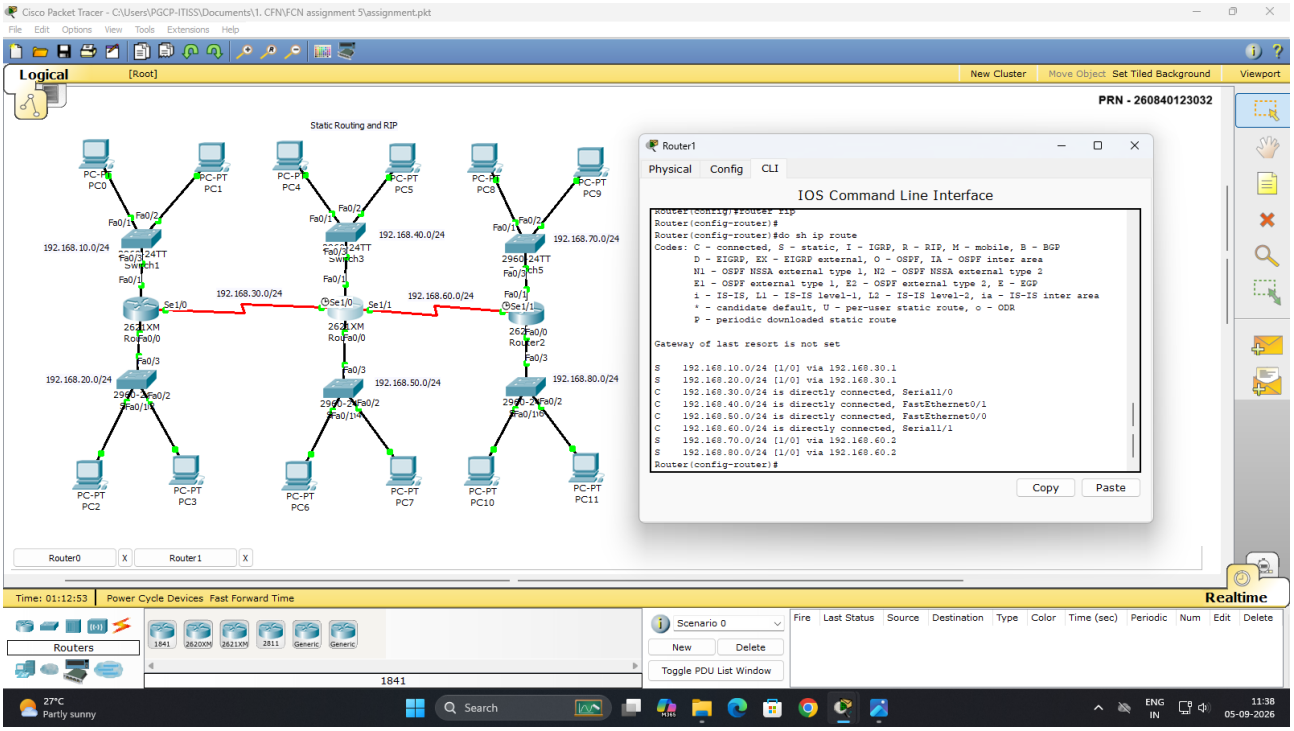
FCN Assignment - 2

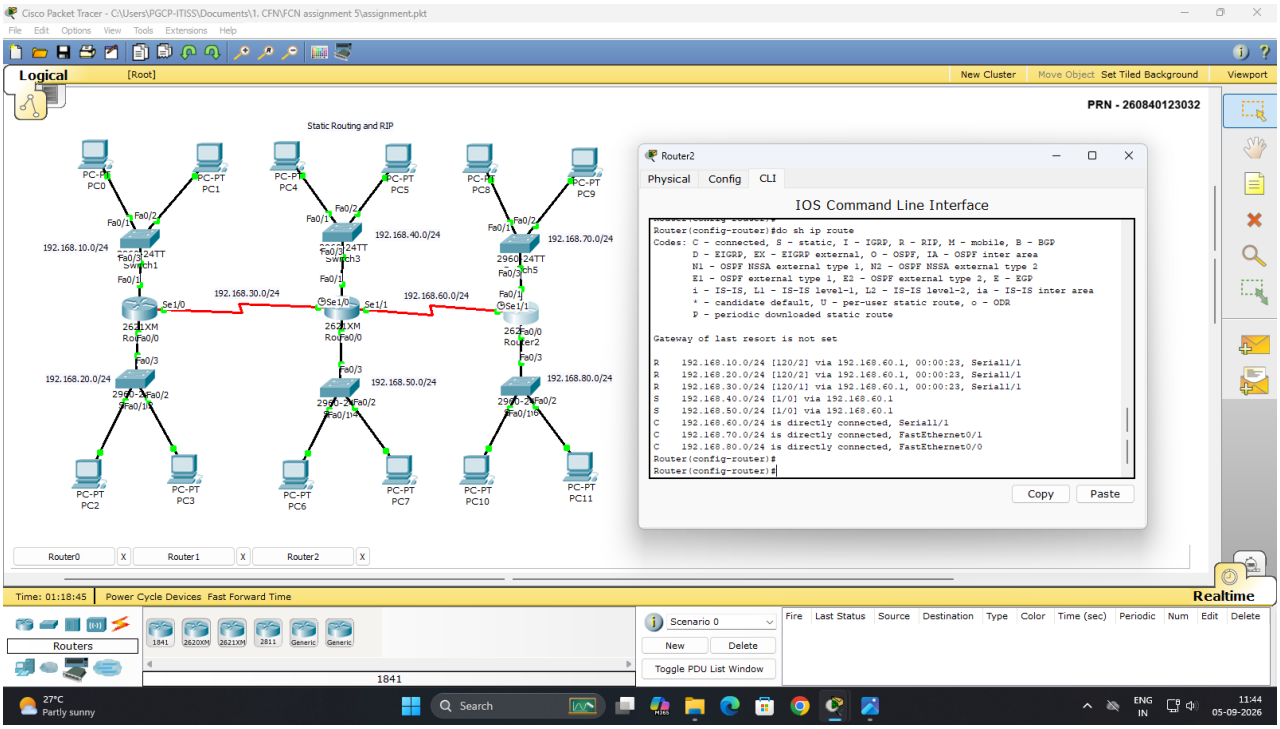
Question 1

Three routers are connected in a line; each router has two switches, and each switch has two PCs. Configure **Static Routing** and **RIP** so that all PCs can communicate.



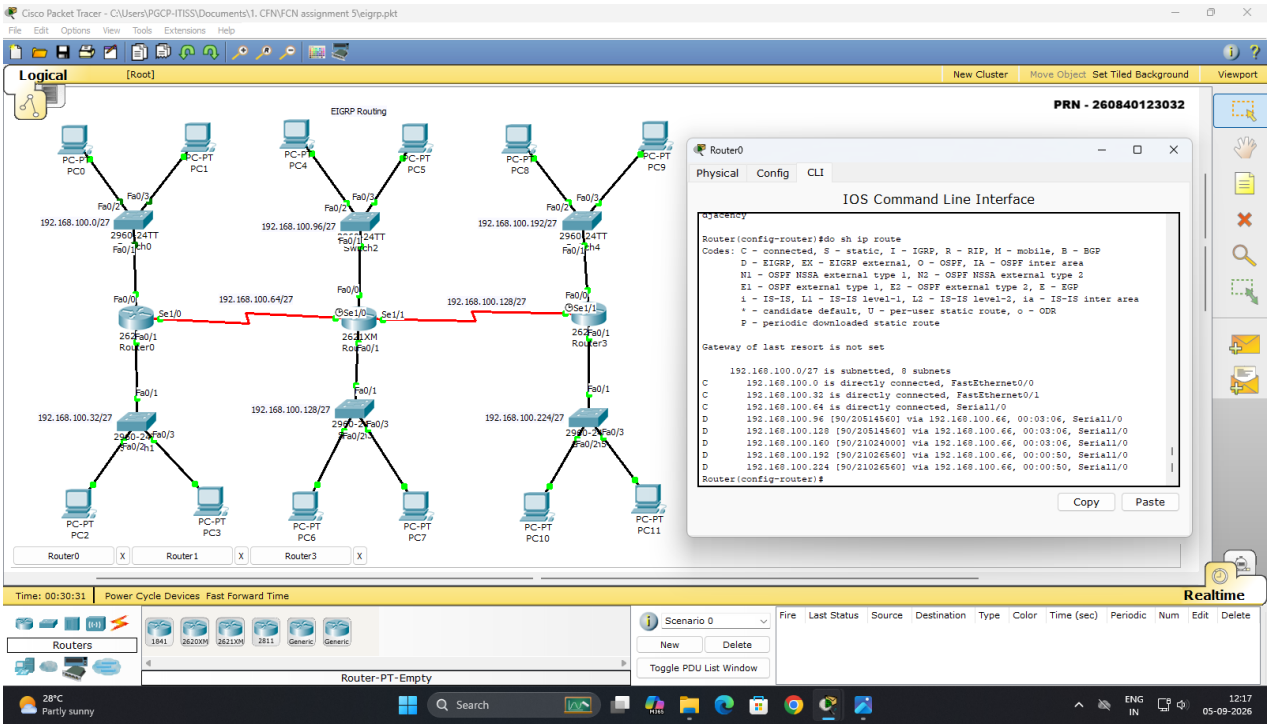


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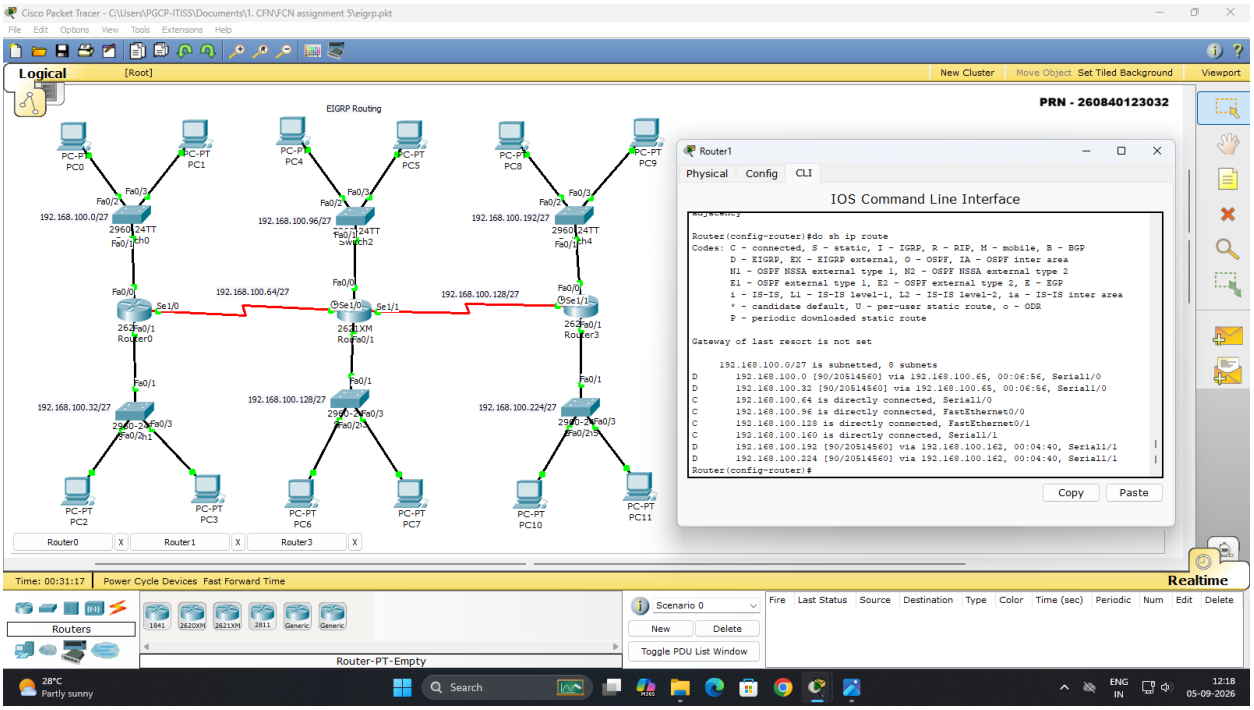


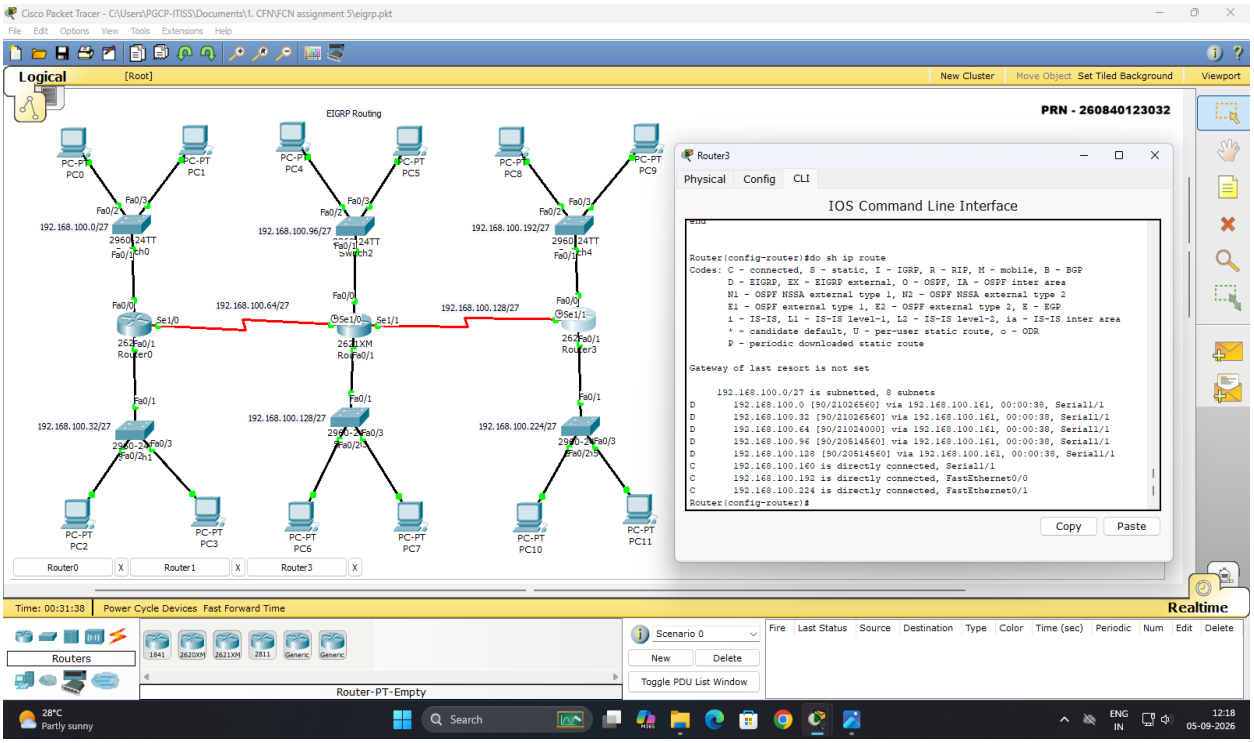
Question 2

Three routers are connected in a line; each router has two switches, and each switch has two PCs. Configure **EIGRP Routing**. Assign the same AS number to the first two routers, and a different AS number to the third router. Check whether the PCs can communicate.



FCN Assignment - 2

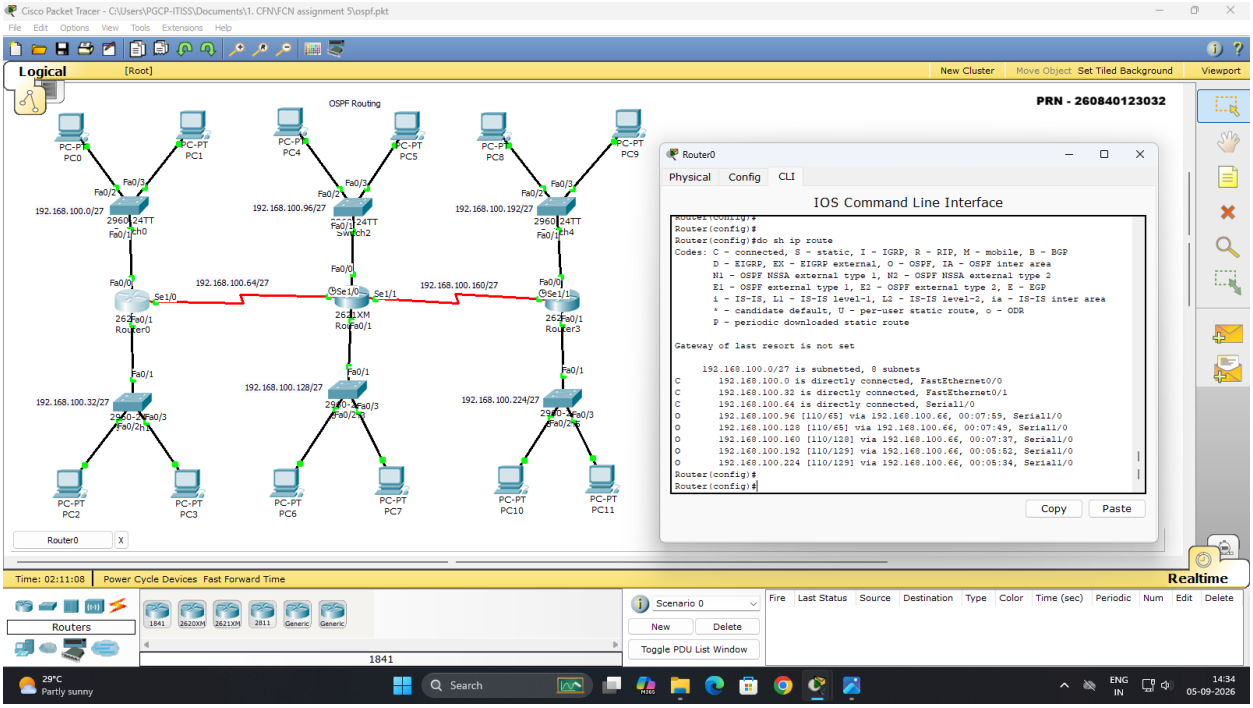


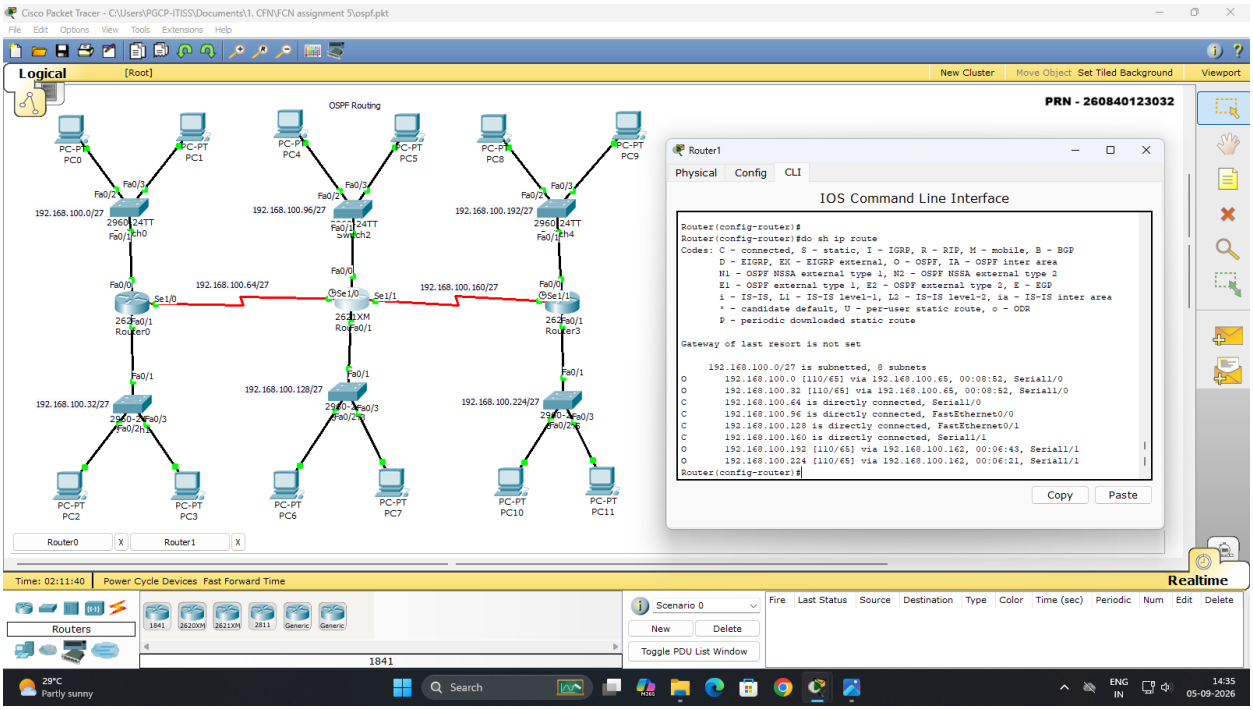


FCN Assignment - 2

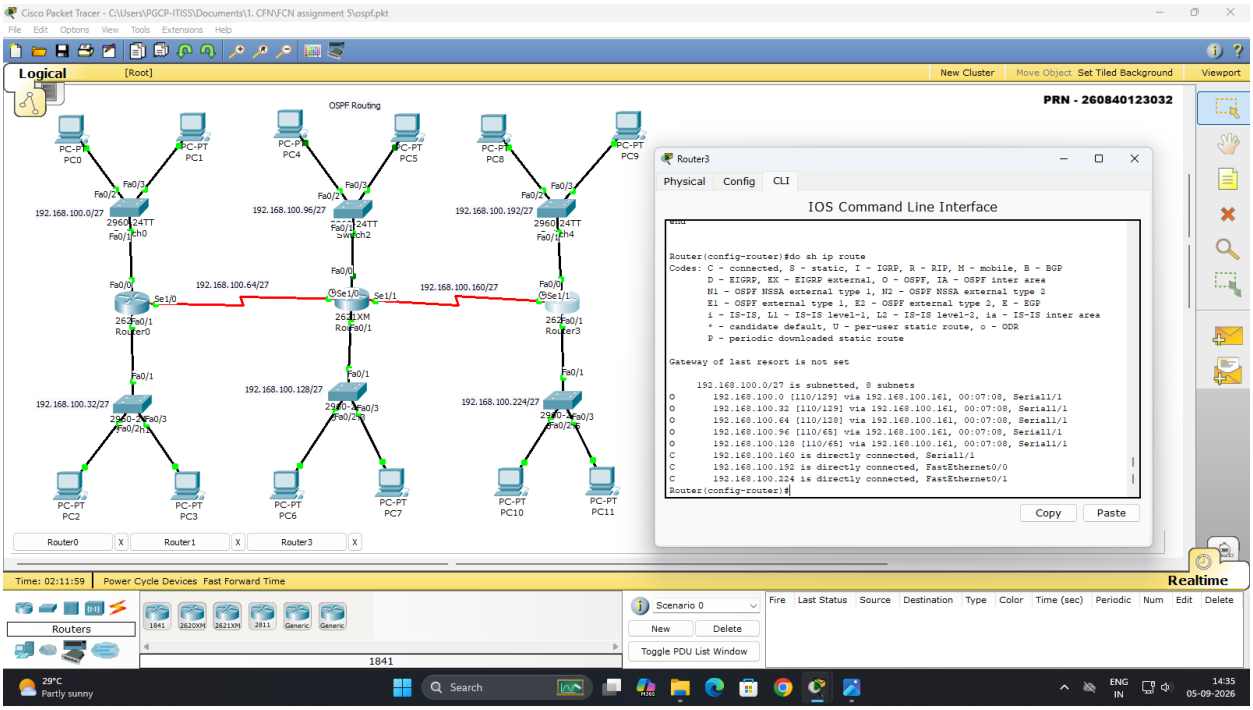
Question 3

Two routers are connected in a line; each router has two switches, and each switch has two PCs. Configure **OSPF**. Make the serial interface a backbone area and the FastEthernet ports non-backbone areas with different area numbers. Check whether the PCs can communicate.





FCN Assignment - 2



Question 4: Boot Process ?

→ The BOOT process is an important concept in computer networks. BOOT stands for Bootstrap Protocol. It is used to help a computer obtain the necessary network information when it starts up. This allows the computer to communicate with other devices on the network.

When a computer is connected to a network, it needs information such as its IP address, subnet mask, default gateway, and sometimes the address of a boot file. The BOOT process helps provide this information, especially when a computer does not already have a configured IP address.

How the BOOT Process Works

The BOOT process takes place when a computer starts and needs to obtain its network configuration. The main steps are:

1. **Computer starts:** The computer boots up and its network interface becomes active.
2. **Request is sent:** The computer sends a BOOTP request to the network to ask for configuration information.
3. **Server receives the request:** A BOOTP server receives the request and checks the computer's hardware address.
4. **Configuration is provided:** The server sends back the required network information, such as the IP address, subnet mask, and default gateway.
5. **Network connection is established:** The computer uses this information to communicate on the network.

BOOTP Message Exchange

The BOOTP process generally involves two main messages:

- BOOTP Request: Sent by the client to request network configuration.
- BOOTP Reply: Sent by the server containing the required configuration information.

By communicating with a BOOTP server, the computer can configure its network connection and begin communicating with other devices. Although DHCP is more commonly used today, BOOTP remains an important concept for understanding network initialization.
In short: BOOTP helps a computer get its network configuration when it starts, so it can communicate on the network.

Question 5: Memory in Router ?

-> A **router** is a networking device that connects different networks and forwards data packets from one network to another. To perform this work efficiently, a router uses different types of memory. Each type of memory has a specific purpose, such as storing the router's operating system, configuration, or information about network routes.

Types of Memory in a Router

1. **RAM (Random Access Memory)**

RAM is the router's working memory. It stores the running configuration, routing tables, ARP cache, and other information needed while the router is operating. RAM is volatile, which means its contents are lost when the router is powered off.

2. **ROM (Read-Only Memory)**

ROM stores the bootstrap program and basic startup instructions. When the router is switched on, it uses ROM to begin the boot process and perform initial hardware checks. ROM is non-volatile, so its contents remain even when the power is off.

3. **Flash Memory**

Flash memory stores the router's operating system, commonly called the IOS in Cisco routers. It is non-volatile and can be updated when a new version of the operating system is installed.

4. **NVRAM (Non-Volatile RAM)**

NVRAM stores the startup configuration of the router. This configuration contains settings such as interface addresses, routing protocols, and passwords. It is saved even when the router is turned off.

FCN Assignment - 2

Difference Between Router Memories

Memory	Main purpose	Volatile?
RAM	Stores information used while the router is running	Yes
ROM	Stores the bootstrap program	No
Flash	Stores the operating system	No
NVRAM	Stores the startup configuration	No

Memory plays an important role in the operation of a router. RAM helps the router work while it is running, ROM helps it start, Flash stores the operating system, and NVRAM stores the configuration needed after a restart. Together, these memories allow the router to start correctly, maintain network information, and forward data efficiently.

Question 6: Administrative Distance (AD) ?

-> **Administrative Distance (AD)** is a value used by routers to decide which routing information is more reliable when they learn the same network through different routing protocols.
In a computer network, a router can learn routes from different sources, such as RIP, OSPF, EIGRP, or a manually configured static route. Sometimes, two or more routing protocols may provide a route to the same destination. In this situation, the router uses Administrative Distance to select the most trustworthy route.
The **lower the Administrative Distance, the more reliable the routing source** is considered.
For example, if a router learns the same route from RIP with an AD of 120 and from OSPF with an AD of 110, the router will choose the OSPF route because 110 is lower than 120.

Common Administrative Distance Values

Routing Source	Administrative Distance
Connected Network	0
Static Route	1
EIGRP	90
OSPF	110
RIP	120
External EIGRP	170

Importance of Administrative Distance

Administrative Distance helps routers choose the best source of routing information when multiple routing protocols are available. It prevents confusion by giving preference to the routing source with the lowest AD value.

For example, if a network administrator configures a static route and the router also learns the same route through OSPF, the static route is normally preferred because its AD is **1**, while OSPF has an AD of **110**.

Administrative Distance is an important concept in routing because it helps a router determine which routing source should be trusted when multiple sources provide information about the same destination. A **lower AD value means higher preference and reliability**. This allows routers to make better decisions and maintain efficient communication within a network.